Perception with Zed X

Carson Fujita

Introduction to Perception

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Camera Specifications

The Zed X is an enviromentally sealed stereo camera using complementary metal oxide semicoductor chip (CMOS) sensors[1][2]. As shown in Table 1, the Zedx CMOS has a RGB color sensor with an array size of 1928 x 1208 photodiodes (2.3MP). It should be safe to assume that the Zed X is a single chip RGB. Unfornately, neither the datasheet or website has any mention of whether this is a 3 chip color camera or not. However, the datasheet mentions that images are provided in RGB to host so it should be safe to assume that this array is a one chip system (StereoLabs, 2023, p.4-5).

[1] StereoLabs, 2023.

[2] "Neousys Enhances GMSL2 Vision with Stereolabs ZED X and ZED X One GS Cameras", 2025, paragraph 1.

Table 1: Zed X Camera Specifications

Sensor Type	CMOS [a]
Array Size	1928 x 1208 px
Pixel Size	3 x 3 micrometers
Shutter	Electronic Synchronized global shutter
Output Format	RAW10
Max SN Ratio	38 dB
Dynamic Range	71.4dB
Responsivity	22.3KeLux-sec
Baseline	12cm

[a] Data from technical specification (StereoLabs, 2023).

[a] Specifically retrieved from Neousys Technology article ("Neousys Enhances GMSL2 Vision with Stereolabs ZED X and ZED X One GS Cameras", 2025).

Understanding CMOS

There a few steps to a CMOS digital camera function:

1. accumulationintergration
2. data collection
3. reading

First, photons hit a 2D array of photodiodes–in this case a 1928 x 1208 array–this exposure accumulates a charge that will be collected. Then, data is collected through pixel-specific circuitry in parallel and amplified. It's important to note that these diodes are slightly more senstive to the infrared end of the visible light spectrum having a range of 400-1000 nm (Siegwart et al., 2011, p. 143). Finally, this data is read by an imaging chip.

Camera configurations change the result of this process. For example, the duration of this period and

amount of light exposed to the diodes are critical to the results; it's important to regulate the amount of photons. When too much light is processed in the collection step this causes blooming where the saturation of the light in the collection causes light to leak to adject pixels in it's array. modifying the intergration and collection. This can be done by changing the aperture (iris position) and shutter speed to modify the integration phase and collection phase respectively (Siegwart et al., 2011, p.143-145)
For color, every 2 by 2 pixels are sorted to read 2 green (the other green pixel is to measure a luminance signal), 1 blue, and 1 red signal. These signals are processing in a process called demosaicing. To control color, we use white balence which changes the measurements of rgb to keep colours consistant in different lightings (Siegwart et al., 2011, p.146-147)

Stereo Vision